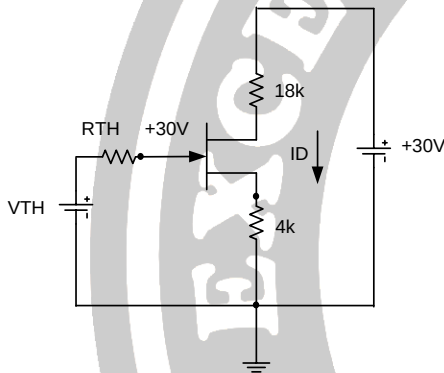




## Answers & Solutions:

1. **C. transconductance,  $g_m$ .**
2. **B. constant current region**
3. **D. can operate with positive as well as negative gate voltages**
4. **D. All of the above**
5. **A. holes**
6. **D. 5 V**
7. **B. normally ON device.**
8. **A. 0.25 mA**

Solution:



$$R_{TH} = 200 \text{ k}\Omega // 1.3 \text{ M}\Omega$$

$$R_{TH} = 173.5 \text{ k}\Omega$$

$$V_{TH} = \frac{200 \text{ k}\Omega}{1500 \text{ k}\Omega} (30) = 4 \text{ V}$$

$$I_D = \frac{V_{TH} - V_{GS}}{4} = \frac{4 - V_{GS}}{4}$$

$$I_D = I_{DSS} \left( \frac{1 - V_{GS}}{V_p} \right)^2$$

$$I_D = \frac{4 - V_{GS}}{4}$$

$$I_D = 4 \left( \frac{1 - V_{GS}}{V_p} \right)^2$$

Solving, with  $V_{GS} = 3 \text{ V}$   
( $V_{GS} = 4$  is not realistic)

$$I_D = \left( 1 - \frac{3}{4} \right)$$

$$I_D = 0.25 \text{ mA} \rightarrow \text{Ans.}$$

9. **C. remains constant**

10. **A.  $V_{GS} > V_p$ .**

11. **A. 1.05 M $\Omega$**

Solution:

$$Z_{in} = R_1 // R_2$$

$$Z_{in} = \frac{R_1 R_2}{R_1 + R_2}$$

$$Z_{in} = \frac{(3.5 \text{ M}\Omega)(1.5 \text{ M}\Omega)}{3.5 \text{ M}\Omega + 1.5 \text{ M}\Omega}$$

$$Z_{in} = 1.05 \text{ M}\Omega \rightarrow \text{Ans.}$$

12. **C. is increased**

13. **D. Both A and B.**

14. **A. gate bias.**

15. **D. The common-drain amplifier.**

16. **A.  $\frac{\Delta V_{DS}}{\Delta I_D}$  at constant  $V_{GS}$**

17. **C. grid cut off voltage**

18. **A. voltage**

19. **D. All of the above.**

20. **B. The common-source amplifier.**

21. **A. 1.261 V**

Solution:

$$I_D = k(V_{GS(ON)} - V_T)^2$$

$$\text{and } (V_{GS(ON)} - V_T)^2 = \frac{I_D}{k}$$

$$V_{GS(ON)} - V_T = \sqrt{\frac{I_D}{k}}$$

$$4\text{V} - V_T = \sqrt{\frac{3 \text{ mA}}{0.4 \times 10^{-3}}}$$

$$V_T = 1.261 \text{ V} \rightarrow \text{Ans.}$$

22. **B. pinch-off voltage,  $V_p$ .**

23. **B.  $I_{DSS}$ .**

24. **B. more than**

25. **A. higher than that of a JET.**

26. **A. normally OFF device.**

27. **A. 0.5 to 30 m A/V**

28. **B. symmetrical JFET.**

29. **C. Voltage controlled current source**

30. **B. source resistor**

31. **A. common drain**

32. **C. Mid- point**

33. **A. The common-gate amplifier.**

34. **B. is decreased**

35. **A. -4.67 V**

Solution:

$$I_D = I_{DSS} \left( 1 - \frac{V_{GS}}{V_p} \right)^2$$

$$14 \text{ mA} = 9.5 \text{ mA} \left( 1 - \frac{+1}{V_p} \right)^2$$

$$V_p = -4.67 \text{ V} \rightarrow \text{Ans.}$$

36. **C. source, gate, drain**

37. **C. is increased**

38. **C. with both positive and negative gate voltage**

39. **C. minimum gate-source voltage that makes drain current flow.**

40. **B. unipolar**

41. **C.  $I_D$  increases in direct proportion to  $V_{DS}$ .**

$$42. \text{ B. } -2 \cdot \frac{I_{DSS}}{V_p} \left( 1 - \frac{V_{GS}}{V_p} \right)$$

43. **B. reverse**

44. **C. The source follower.**



45. **C. Pentode**

46. **A. normally ON device.**

47. **A.**  $I_D = I_{DSS} \left( 1 - \frac{V_{GS}}{V_p} \right)^2$

48. **B. 11.11 mA**

Solution:

$$I_D = I_{DSS} \left( 1 - \frac{V_{GS}}{V_p} \right)^2$$

$$I_{DSS} = \frac{I_D}{\left( 1 - \frac{V_{GS}}{V_p} \right)^2} = \frac{4 \text{ mA}}{\left( 1 - \frac{(-2V)}{(-5V)} \right)^2}$$

$$I_{DSS} = 11.11 \text{ mA} \rightarrow \text{Ans.}$$

49. **A. voltage**

50. **D. 5 V**

END